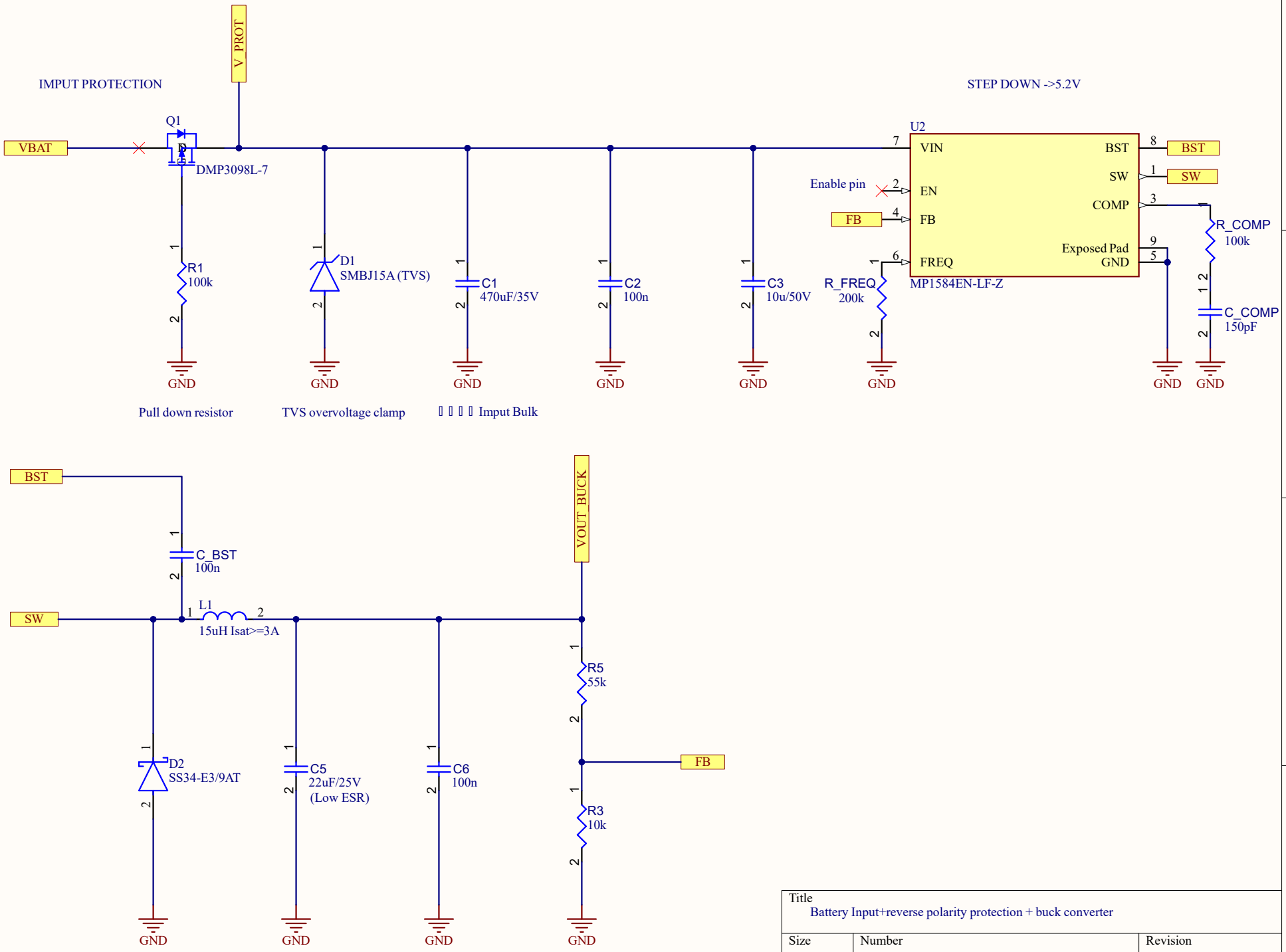
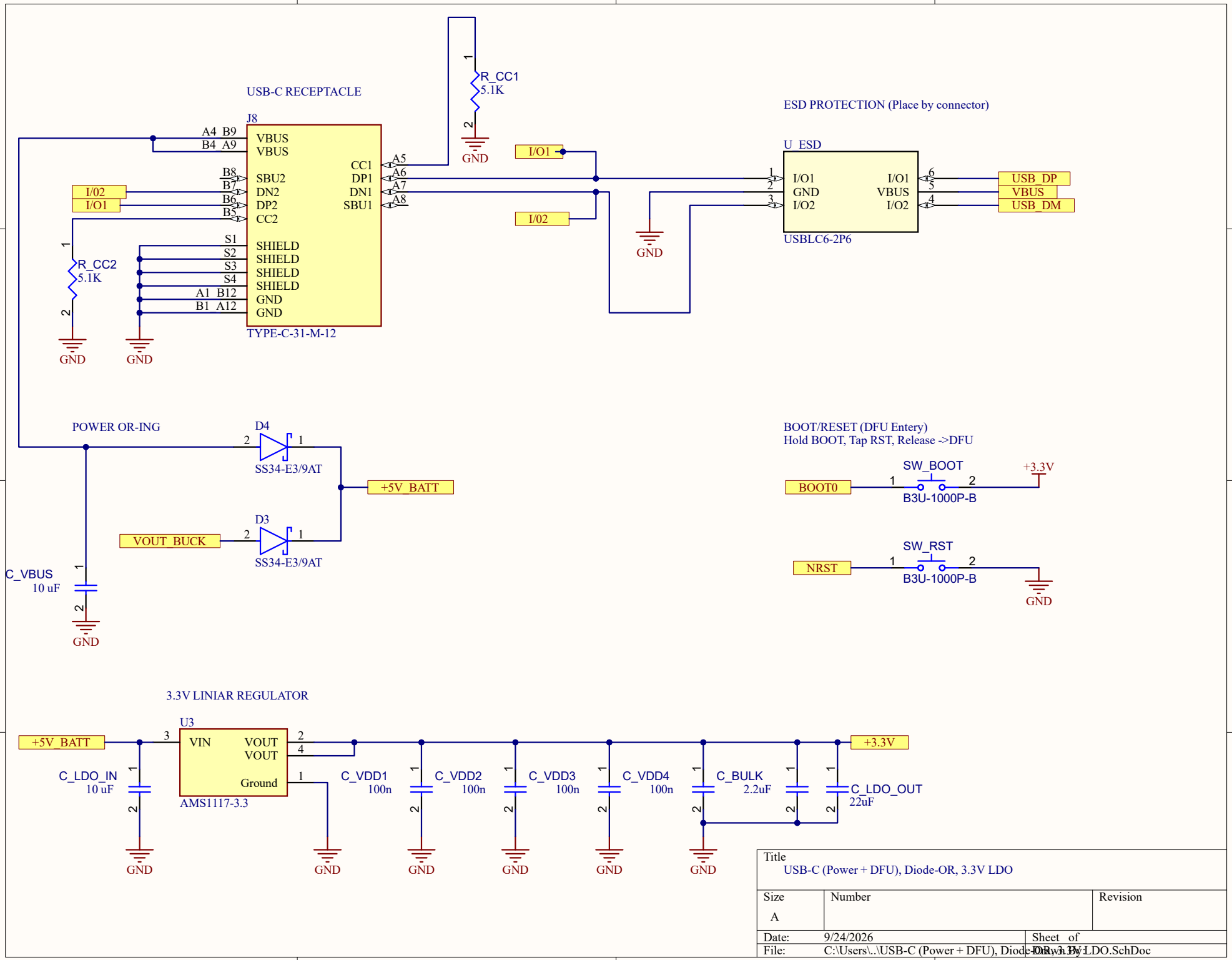
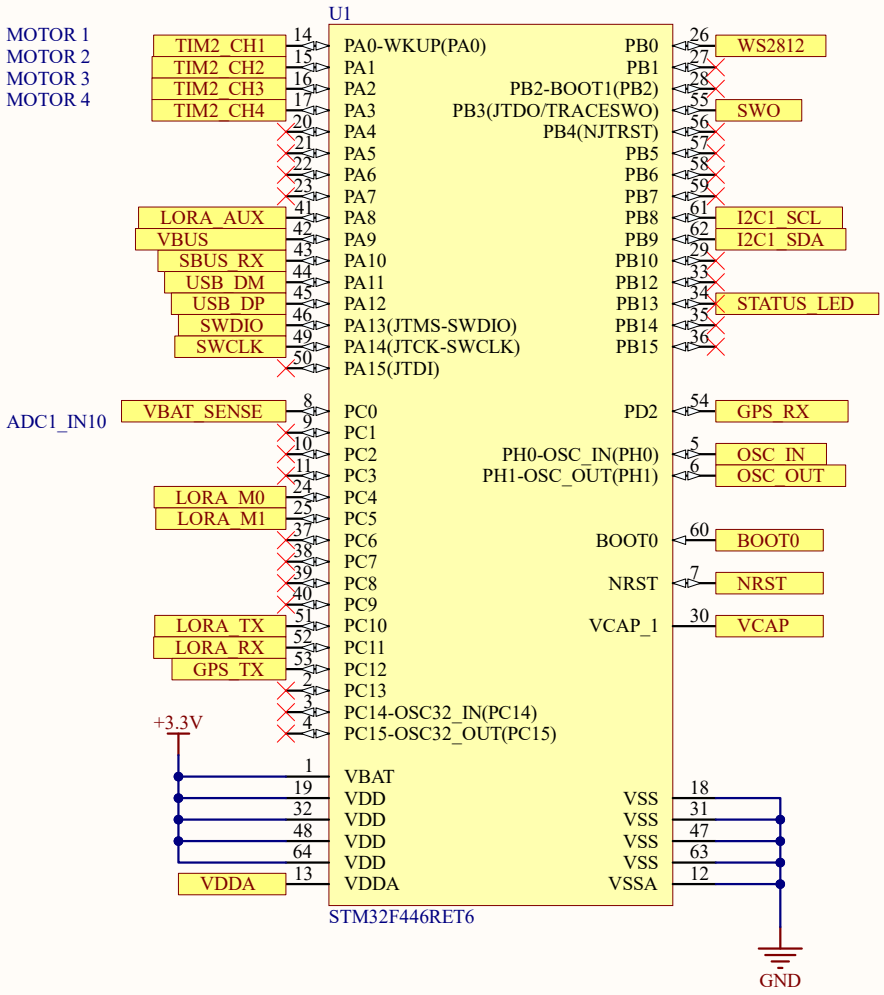
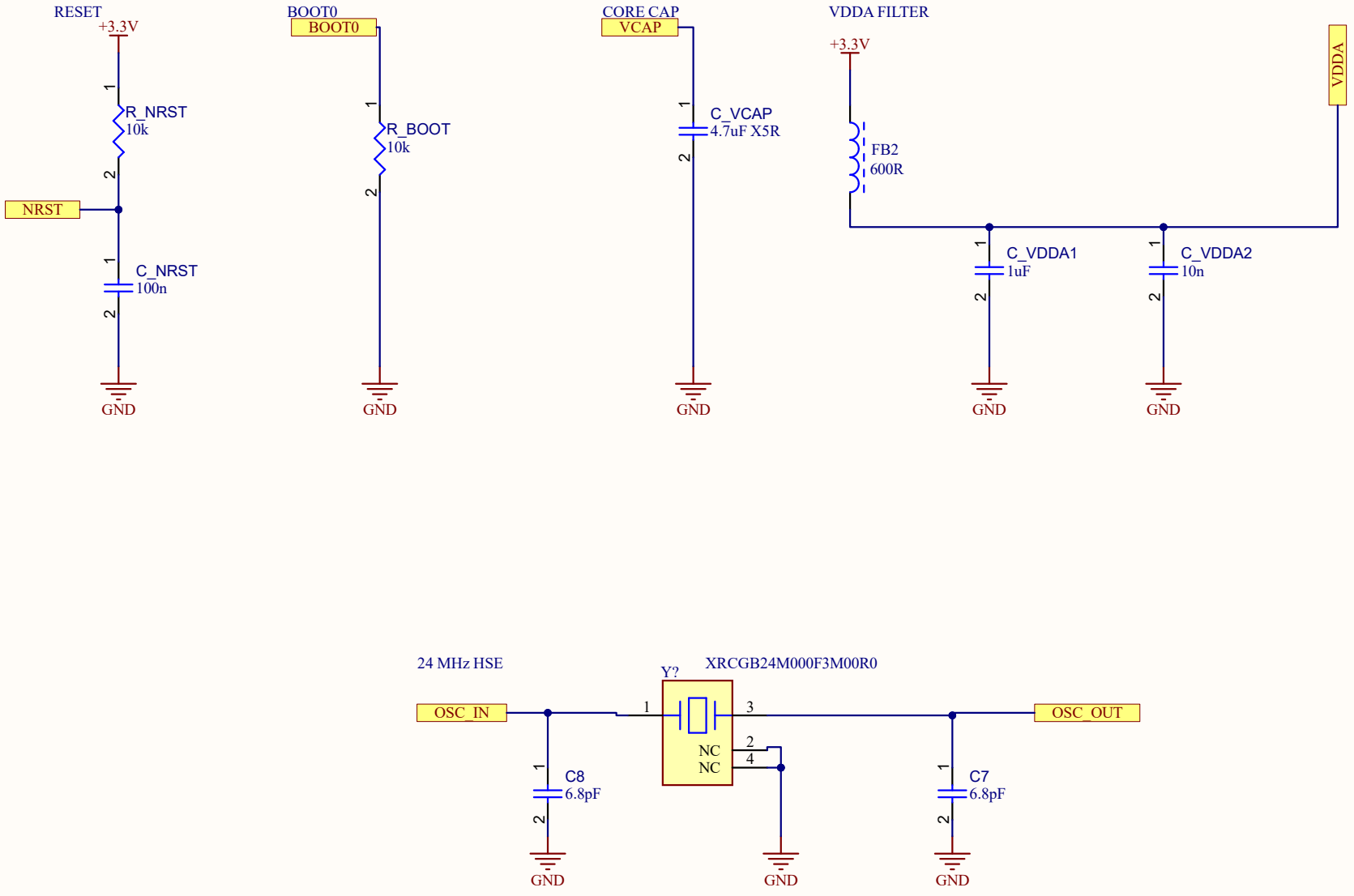

IMPUT PROTECTION
V_PROT
VBAT
Q1
DMP3098L-7
R1
100k
D1
SMBJ15A (TVS)
C1
470uF/35V
C2
100n
C3
10u/50V
GND
Pull down resistor
TVS overvoltage clamp
Imput Bulk
STEP DOWN ->5.2V
U2
VIN
EN
FB
FREQ
BST
SW
COMP
Exposed Pad
GND
MP1584EN-LF-Z
Enable pin
FB
R_FREQ
200k
R_COMP
100k
C_COMP
150pF
BST
SW
C_BST
100n
L1
15uH Isat>=3A
VOUT_BUCK
D2
SS34-E3/9AT
C5
22uF/25V
(Low ESR)
C6
100n
R5
55k
R3
10k
FB
Title
Battery Input+reverse polarity protection + buck converter
Size
A
Number
Revision
Date:
9/24/2026
Sheet of
File:
C:\Users\..\Battery Input, Protection, 5.2V Buck.SchDoc
Drawn By:

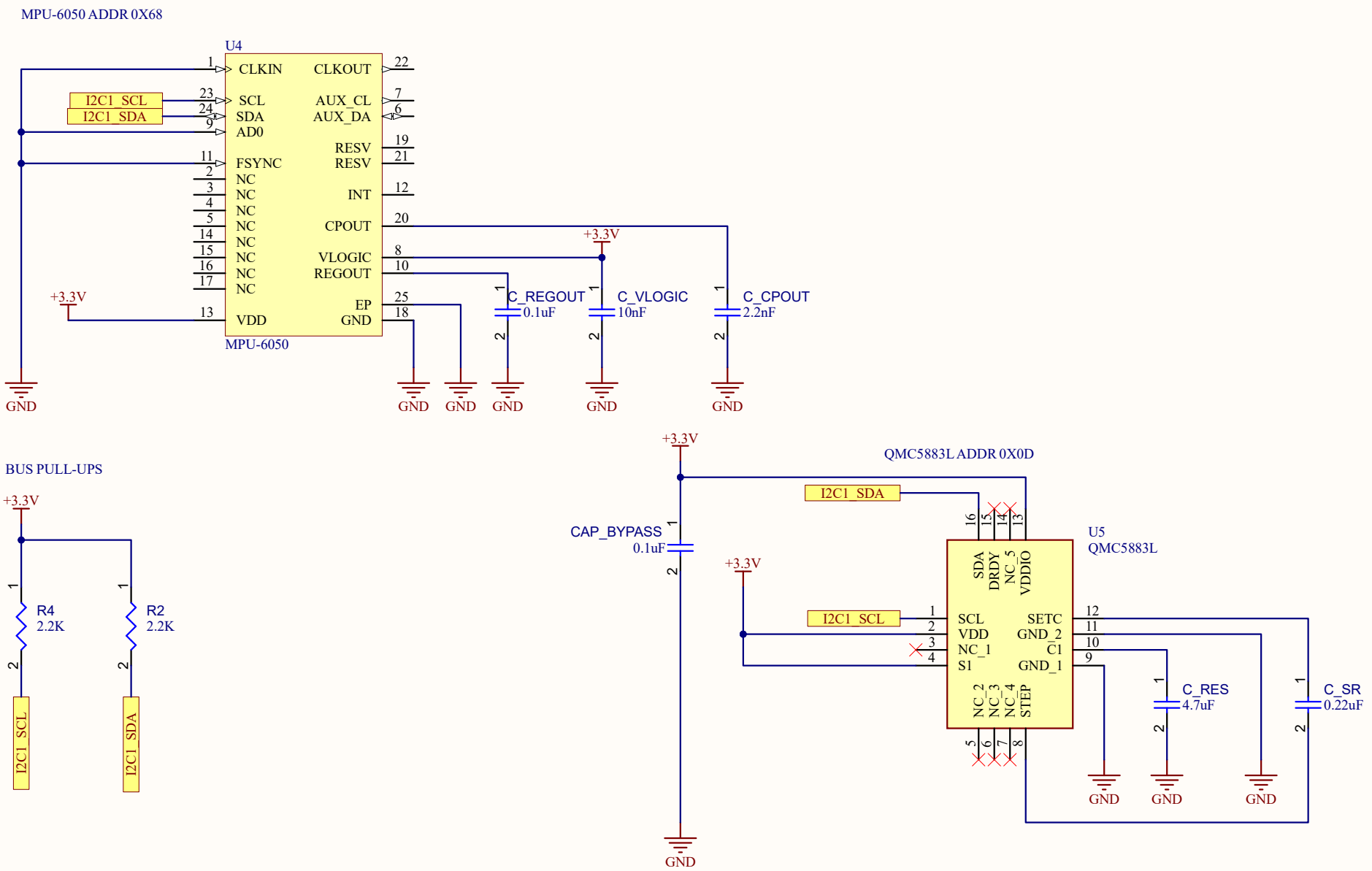

USB-C RECEPTACLE
J8
A4 B9
B4 A9
VBUS
VBUS
B8
B7
B6
B5
SBU2
DN2
DP2
CC2
CC1
DP1
DN1
SBU1
A5
A6
A7
A8
S1
S2
S3
S4
SHIELD
SHIELD
SHIELD
SHIELD
A1 B12
B1 A12
GND
GND
TYPE-C-31-M-12
R_CC1
5.1K
R_CC2
5.1K
I/02
I/O1
GND
ESD PROTECTION (Place by connector)
U_ESD
I/O1
GND
I/O2
I/O1
VBUS
I/O2
USBLC6-2P6
USB_DP
VBUS
USB_DM
POWER OR-ING
D4
SS34-E3/9AT
D3
SS34-E3/9AT
+5V_BATT
VOUT_BUCK
C_VBUS
10 uF
BOOT/RESET (DFU Entery)
Hold BOOT, Tap RST, Release ->DFU
SW_BOOT
BOOT0
+3.3V
B3U-1000P-B
SW_RST
NRST
B3U-1000P-B
3.3V LINIAR REGULATOR
U3
+5V_BATT
VIN
VOUT
VOUT
Ground
AMS1117-3.3
C_LDO_IN
10 uF
C_VDD1
100n
C_VDD2
100n
C_VDD3
100n
C_VDD4
100n
C_BULK
2.2uF
C_LDO_OUT
22uF
+3.3V
Title
USB-C (Power + DFU), Diode-OR, 3.3V LDO
Size
A
Number
Revision
Date:
9/24/2026
Sheet of
File:
C:\Users\..\USB-C (Power + DFU), Diode-OR, 3.3V LDO.SchDoc
Drawn By:



Title: STM32F446RET6 Core		
Size: A	Number:	Revision:
Date: 9/24/2026		Sheet of
File: C:\Users\..\STM32F446RET6 Core.SchDoc		Drawn By:

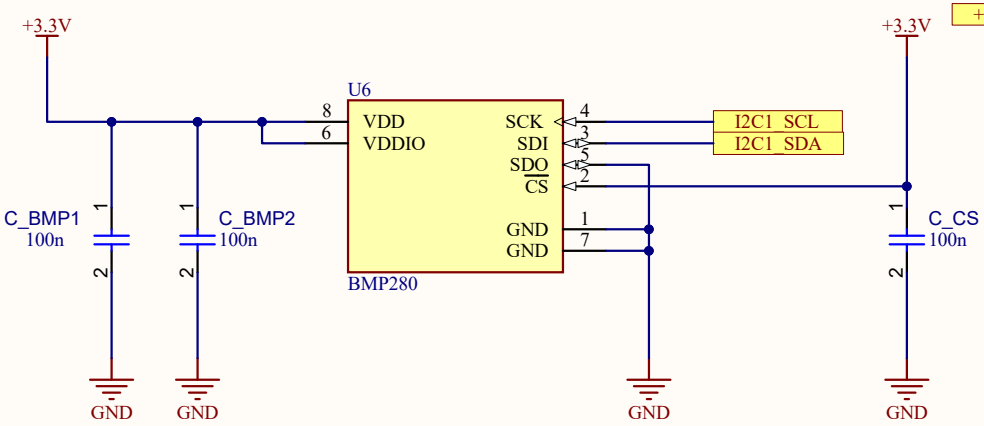


Title: MCU Support Circuits		
Size: A	Number:	Revision:
Date: 9/24/2026		Sheet of
File: C:\Users\..\MCU Support Circuit.SchDoc		Drawn By:

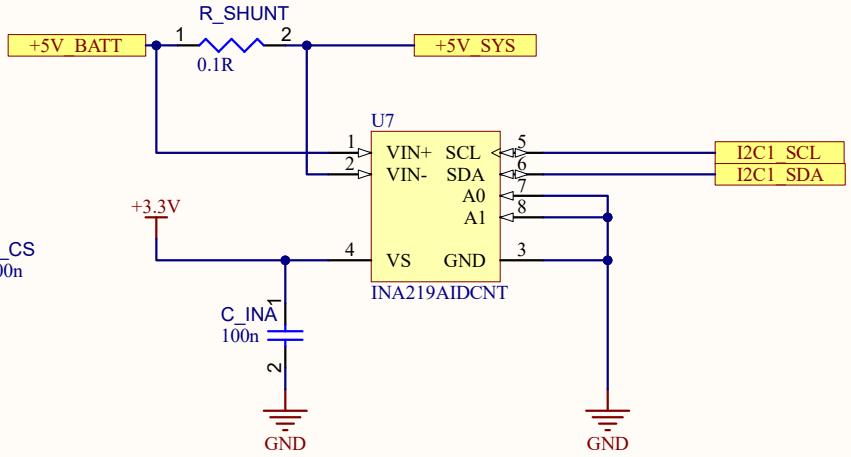


Title: I2C Sensors A- IMU and Magnetometer		
Size: A	Number:	Revision:
Date: 9/24/2026		Sheet of
File: C:\Users\..\I2C Sensors A- IMU and Magnetometer.SchDoc		Drawn By:

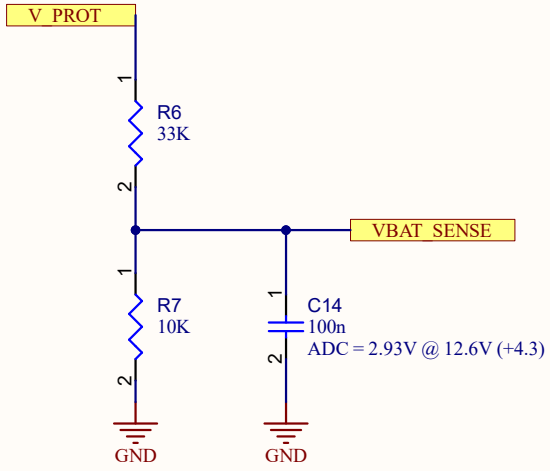
BMP 280 ADDR 0X76



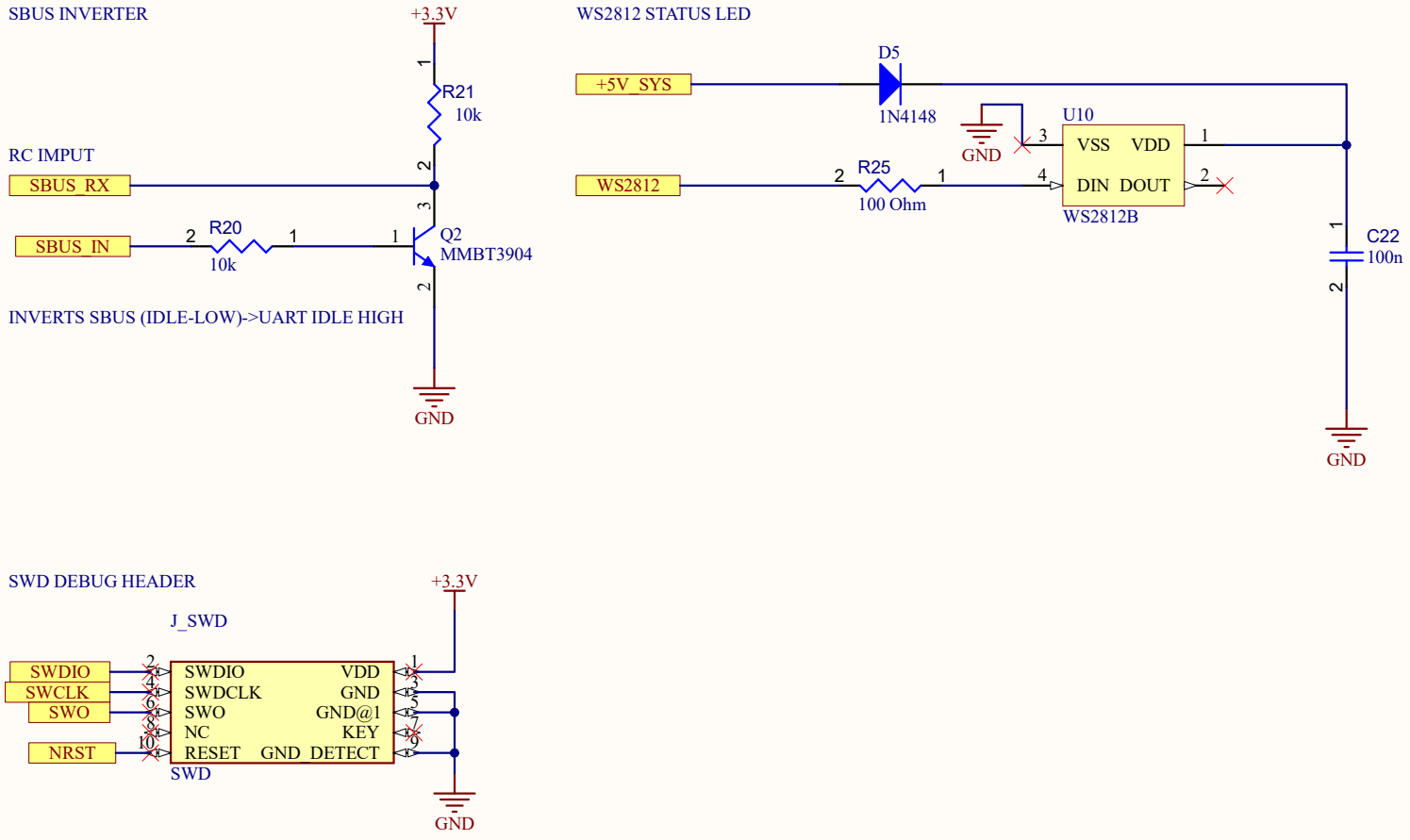
INA219 + SHUNTADDR 0X40



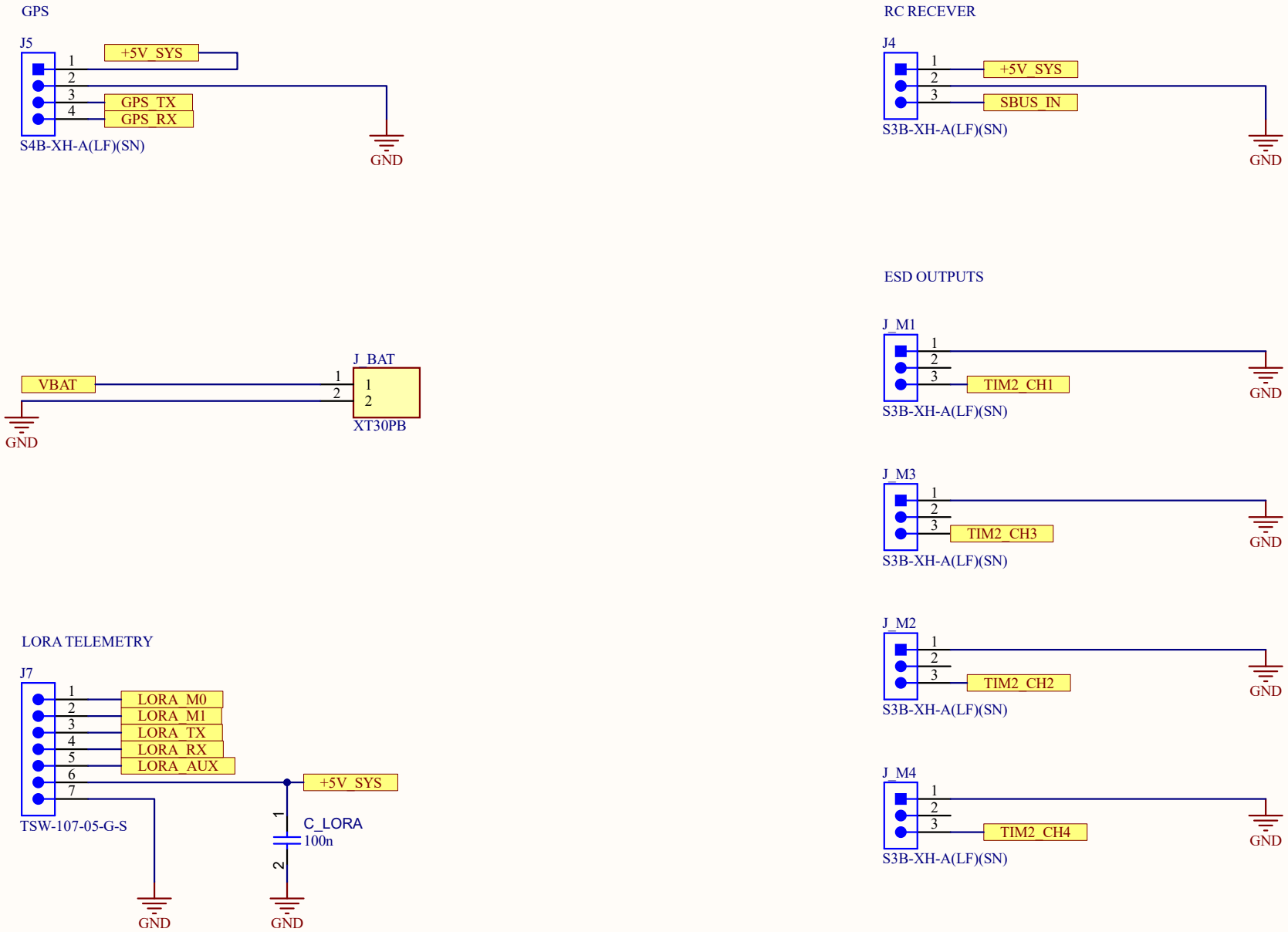
BATTERY SENSE



Title: I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE		
Size: A	Number:	Revision:
Date: 9/24/2026		Sheet of
File: C:\Users\..\I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE.		Drawn By:



Title: RC INPUT, STATUS LED, SWD HEADER		
Size: A	Number:	Revision:
Date: 9/24/2026		Sheet of
File: C:\Users\..\RC INPUT, STATUS LED, SWD HEADER.SchDoc		Drawn By:


GPS
J5
+5V_SYS
GPS_TX
GPS_RX
S4B-XH-A(LF)(SN)
GND
RC RECEVER
J4
+5V_SYS
SBUS_IN
S3B-XH-A(LF)(SN)
GND
ESD OUTPUTS
J_M1
TIM2_CH1
S3B-XH-A(LF)(SN)
GND
J_M3
TIM2_CH3
S3B-XH-A(LF)(SN)
GND
J_M2
TIM2_CH2
S3B-XH-A(LF)(SN)
GND
J_M4
TIM2_CH4
S3B-XH-A(LF)(SN)
GND
VBAT
J_BAT
XT30PB
GND
LORA TELEMETRY
J7
LORA_M0
LORA_M1
LORA_TX
LORA_RX
LORA_AUX
+5V_SYS
TSW-107-05-G-S
C_LORA
100n
GND
GND
Title
EXTERNAL CONNECTORS
Size
A
Number
Revision
Date:
9/24/2026
Sheet of
File:
C:\Users\..\EXTERNAL CONNECTORS.SchDoc
Drawn By: